

Gastric acid is one of many acids important to biological function in the human body. Production of the main component of gastric acid, hydrochloric acid (HCl), is accomplished in the stomach by the parietal cells, as shown in Figure 1.

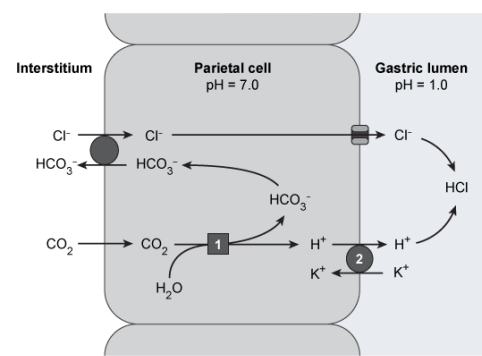


Figure 1 Formation of gastric acid in the stomach

Facilitated by the carbonate dehydratase enzyme (1), the parietal cells utilize the interaction between aqueous carbon dioxide and water in the blood as a source of bicarbonate ions and acidic H⁺ ions (protons). The H⁺ ions produced in this process are then exchanged for potassium ions through the cell membrane by a proton pump, the H⁺/K⁺-ATPase (2). In conjunction with this proton exchange, bicarbonate ions are also exchanged for chloride ions from the interstitial fluid. As a result, H⁺ and Cl⁻ ions are concentrated in the lumen of the stomach to form aqueous hydrochloric acid.

Because the production of the acid proceeds with a drop in pH from 7.0 in the cell cytosol to 1.0 in the stomach lumen, the process occurs against a concentration gradient and is endergonic, requiring the input of energy supplied by the hydrolysis of ATP. Ignoring effects from ion charge, the free energy change ΔG_t for ion transport across the cell membrane can be estimated by Equation 1:

$$\Delta G_t = kRT \log\left(\frac{C_f}{C_i}\right)$$

Equation 1

where R is the gas constant (8.32 J/mol·K), k is a logarithm conversion constant, T is the absolute temperature, and C_i and C_f are the initial and final molar concentrations across the gradient, respectively.

Although dilute, the gastric (hydrochloric) acid released into the lumen yields a strongly acidic environment crucial to protein digestion and the destruction of harmful micro-organisms. A comparison of its acid dissociation constant K_a to those of other biologically relevant acids is given in Table 1.

Table 1 Comparison of Strengths of Selected 0.10 M Acids at 25°C

Acid	Acid Formula (HA)	Conjugate Base (A ⁻)	K_a
Hydrochloric	HCl	Cl ⁻	1.3×10^6
Carbonic	H ₂ CO ₃	HCO ₃ ⁻	4.3×10^{-7}
Formic	HCO ₂ H	CO ₂ H ⁻	1.8×10^{-4}
Acetic	HC ₂ H ₃ O ₂	C ₂ H ₃ O ₂ ⁻	1.8×10^{-5}
Lactic	H ₃ C ₃ H ₅ O ₃	HC ₃ H ₅ O ₃ ⁻	1.4×10^{-4}
Ascorbic	H ₂ C ₆ H ₆ O ₆	HC ₆ H ₆ O ₆ ⁻	7.9×10^{-5}
Citric	H ₃ C ₆ H ₇ O ₇	H ₂ C ₆ H ₇ O ₇ ⁻	8.4×10^{-4}

Based on the acid comparison in Table 1, which of the following acids will give a 0.10 M aqueous solution with the lowest pH?

- ☐ A. Carbonic acid [13%]
☐ B. Acetic acid [1%]
☐ C. Lactic acid [13%]
☒ D. Citric acid [71%]

Correct

71% answered correctly

Explanation:

$HA \rightleftharpoons H^+ + A^-$

$K_a = \frac{[H^+][A^-]}{[HA]}$

$pK_a = -\log(K_a)$

$pH = -\frac{1}{2} \log([HA]K_a)$

Acid	HA	A ⁻	K _a	pK _a	pH
Citric	H ₃ C ₆ H ₅ O ₇	H ₂ C ₆ H ₄ O ₇ ⁻	8.4 × 10 ⁻⁴	3.1	2.04
Formic	HCO ₂ H	CO ₂ H ⁻	1.8 × 10 ⁻⁴	3.7	2.37
Lactic	H ₃ C ₃ H ₅ O ₃	HC ₂ H ₃ O ₂ ⁻	1.4 × 10 ⁻⁴	3.9	2.43
Ascorbic	H ₂ C ₆ H ₇ O ₆	HC ₆ H ₆ O ₆ ⁻	7.9 × 10 ⁻⁵	4.1	2.55
Acetic	HC ₂ H ₃ O ₂	C ₂ H ₃ O ₂ ⁻	1.8 × 10 ⁻⁵	4.7	2.87
Carbonic	H ₂ CO ₃	HCO ₃ ⁻	4.3 × 10 ⁻⁷	6.4	3.68

Increasing acid strength

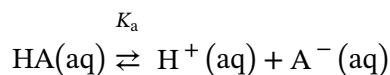
↑

decreasing acid strength

↓

HA
A⁻

When an **acid (HA)** is dissolved and ionized in water, a proton (H⁺ ion) and a **conjugate base** of the acid (A⁻) are released into solution.



For this process, **strong acids** have an equilibrium favoring the products (ie, the formation of H⁺ and A⁻) whereas **weak acids** have an equilibrium favoring the reactant (non-ionized acid, HA). Consequently, stronger acids produce higher concentrations of H⁺ and A⁻ in solution relative to non-ionized HA. This equilibrium is quantified by the **acid ionization constant K_a**, which is **larger for stronger acids**.

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

Due to the wide range in K_a values, the **pK_a** scale is frequently used as a convenient way of comparing acid strength.

$$pK_a = -\log K_a$$

Because of the logarithmic definition of the pK_a scale, **acid strength increases as pK_a decreases**.

Analogously, the **pH of a solution** is defined as:

$$pH = -\log[H^+]$$

Accordingly, the acid that most favors H⁺ production yields the lowest pH. Therefore, when compared in equal amounts, the acid with the **smallest pK_a (largest K_a)** is the **strongest acid** and yields the solution with the **highest [H⁺]** and the **lowest pH**.

Of the acids compared in Table 1, citric acid is the most acidic (ie, largest K_a, smallest pK_a) and will produce a solution with the lowest (**most acidic**) pH.

(Choices A, B, and C) Acetic, carbonic, and lactic acids all have a smaller K_a (larger pK_a) than citric acid, indicating that these are weaker acids. Solutions of these acids result in *higher* pH values (ie, less acidic solutions).

Educational objective:

Acid strength increases as the acid ionization constant K_a increases, or as the acid pK_a decreases. As a result, the pH of a solution at a given acid concentration decreases as the pK_a decreases.

Time spent:0 sec

Subject:
General Chemistry

QID:401571

System:
Solutions and Electrochemistry

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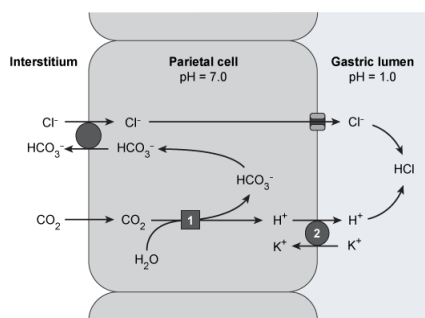


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Lactic	$\text{H}_3\text{C}_3\text{H}_5\text{O}_3$	$\text{HC}_3\text{H}_5\text{O}_3^-$	1.4×10^{-4}
Ascorbic	$\text{H}_2\text{C}_6\text{H}_6\text{O}_6$	$\text{HC}_6\text{H}_6\text{O}_6^-$	7.9×10^{-5}
Citric	$\text{H}_3\text{C}_6\text{H}_7\text{O}_7$	$\text{H}_2\text{C}_6\text{H}_7\text{O}_7^-$	8.4×10^{-4}

Based on the biological system described in the passage, how many millimoles of hydrogen ions are present in 10.0 mL of gastric acid? (Note: Assume that the pH of the acid is equal to the pH of the gastric lumen.)

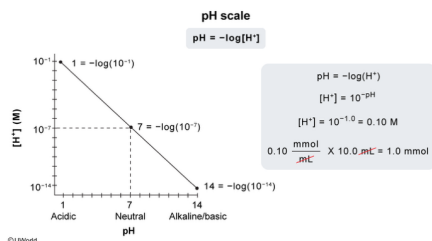
- ☐ A. 0.10 mmol [10%]
☒ B. 1.0 mmol [48%]
☐ C. 10 mmol [23%]

☐ D. 100 mmol [17%]

Correct

48% answered correctly

Explanation:



The **pH scale** defines pH as the negative **logarithm (base 10)** of the **hydrogen ion concentration**:

$$\text{pH} = -\log [\text{H}^+]$$

Rearranging this relationship, the molar concentration of hydrogen ions can be calculated as:

$$[\text{H}^+] = 10^{-\text{pH}}$$

The passage states gastric acid has a pH of 1.0. Assuming the gastric acid has a pH equal to that of the gastric lumen, the concentration of hydrogen ions is:

$$[\text{H}^+] = 10^{-1.0} = 1 \times 10^{-1} = 0.10 \text{ M}$$

The **molar concentration** (ie, molarity) by definition has units of **moles/liter**, but the numerator and denominator can be **scaled by the same factor** to give the equivalent concentration in **millimoles/milliliter** (mmol/mL).

$$0.10 \frac{\text{mol}}{\text{L}} \times \frac{1,000 \text{ mmol}}{1 \text{ mol}} \times \frac{1 \text{ L}}{1,000 \text{ mL}} = 0.10 \frac{\text{mmol}}{\text{mL}}$$
$$0.10 \frac{\text{mol}}{\text{L}} = 0.10 \frac{\text{mmol}}{\text{mL}}$$

Multiplying the volume (mL) of the solution (ie, the gastric acid) by its molar concentration (in mmol/mL) yields the number of **millimoles of acid** solute contained in that volume of the solution. Therefore, the number of mmol of H^+ ions in 10.0 mL of gastric acid is:

$$10.0 \text{ mL} \times 0.10 \frac{\text{mmol H}^+}{\text{mL}} = 1.0 \text{ mmol H}^+$$

(Choice A) 0.10 mmol is too small by a factor of 10. This value may result by mistaking the concentration *molarity* (moles/liter) for moles and failing to multiply by the sample volume.

(Choice C) 10 mmol is too large by a factor of 10. This value may result if the pH value is mistaken as the concentration and then multiplied by the given volume of gastric acid.

(Choice D) 100 mmol is too large by a factor of 100 and may result from **algebraic errors**.

Educational objective:

The pH of a solution is defined as $\text{pH} = -\log[\text{H}^+]$, where $[\text{H}^+]$ represents the hydrogen ion concentration in units of molarity (moles per liter). Knowing the pH of a given volume of solution allows the determination of the molar concentration and the number of moles of hydrogen ions present.

Time spent: 0 sec
Subject:
General Chemistry

QID: 401572
System:
Solutions and Electrochemistry

Gastric acid is one of many acids important to biological function in the human body. Production of the main component of gastric acid, hydrochloric acid (HCl), is accomplished in the stomach by the parietal cells, as shown in Figure 1.

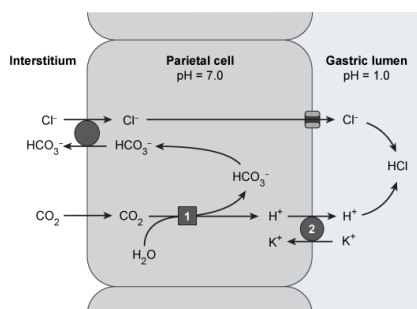


Figure 1 Formation of gastric acid in the stomach

Facilitated by the carbonate dehydratase enzyme (1), the parietal cells utilize the interaction between aqueous carbon dioxide and water in the blood as a source of bicarbonate ions and acidic H^+ ions (protons). The H^+ ions produced in this process are then exchanged for potassium ions through the cell membrane by a proton pump, the H^+/K^+ -ATPase (2). In conjunction with this proton exchange, bicarbonate ions are also exchanged for chloride ions from the interstitial fluid. As a result, H^+ and Cl^- ions are concentrated in the lumen of the stomach to form aqueous hydrochloric acid.

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Equation 1

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Acetic	$HC_2H_3O_2$	$C_2H_3O_2^-$	1.8×10^{-5}
Lactic	$H_3C_3H_5O_3$	$HC_3H_5O_3^-$	1.4×10^{-4}
Ascorbic	$H_2C_6H_6O_6$	$HC_6H_6O_6^-$	7.9×10^{-5}
Citric	$H_3C_6H_7O_7$	$H_2C_6H_7O_7^-$	8.4×10^{-4}

Which of the following expressions gives the estimated change in Gibbs free energy needed to transport H^+ ions across the cell membrane from the parietal cell cytosol to the stomach lumen at physiological temperature?

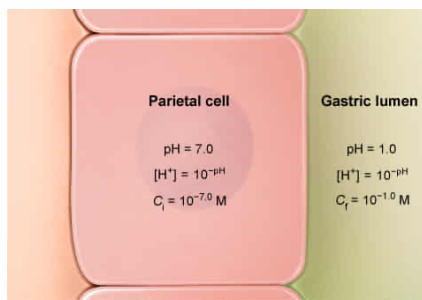
- ☐ A. $\frac{1}{7} kRT \text{ J/mol}$ [23%]
- ✓ ☒ B. $6.0kRT \text{ J/mol}$ [51%]
- ☐ C. $7.0kRT \text{ J/mol}$ [21%]

○ D. $8.0kRT$ J/mol [3%]

Correct

51% answered correctly

Explanation:



$$\Delta G_t = kRT \log \left(\frac{C_f}{C_i} \right)$$

$$\Delta G_t = kRT \log \left(\frac{10^{-1.0} \text{ M}}{10^{-7.0} \text{ M}} \right) = kRT \log(10^{6.0})$$

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$$\Delta G_t = 6.0kRT$$

The observed change in the **Gibbs free energy ΔG** of a system that is **not at equilibrium** can be generally described by the equation:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

where ΔG° is the Gibbs free energy under **standard conditions**, R is the **gas constant**, T is the **Kelvin temperature**, and Q is the **reaction quotient** for the system. As the system reaches **equilibrium**, ΔG goes to zero and Q **adjusts** to become equal to the **equilibrium constant K_{eq}** . Accordingly, the change in the **standard Gibbs free energy ΔG°** of the system **at equilibrium** is given by:

$$0 = \Delta G^\circ + RT \ln K_{eq}$$

$$\Delta G^\circ = -RT \ln K_{eq}$$

In the passage, the gastric system is not at equilibrium. Consequently, Equation 1 in the passage represents a specific case of the ΔG equation where $\Delta G_t = (\Delta G - \Delta G^\circ)$, $Q = C_f/C_i$ (ie, the **ratio of the hydrogen ion concentrations $[H^+]$** on each side of the membrane), and k is a constant to convert from the natural logarithm to the **base 10 logarithm**.

$$\Delta G = \Delta G^\circ + RT \ln Q$$

$$\Delta G - \Delta G^\circ = RT \ln \left(\frac{C_f}{C_i} \right)$$

$$\Delta G_t = kRT \log \left(\frac{C_f}{C_i} \right)$$

Because the H^+ ion concentration correlates directly to **pH** (ie, $[H^+] = 10^{-pH}$), the pH values of the cell cytosol (7.0) and the stomach lumen (1.0) can be used to calculate $[H^+]$ for each. In the gastric system, the hydrogen ions are being concentrated from the cell to the lumen, so the final concentration C_f pertains to the stomach ($C_f = 10^{-1.0} = 1.0 \times 10^{-1}$ M) whereas the initial concentration C_i pertains to the cell ($C_i = 10^{-7.0} = 1.0 \times 10^{-7}$ M). Substituting these values into Equation 1 gives:

$$\Delta G_t = kRT \log \left(\frac{10^{-1.0}}{10^{-7.0}} \right)$$

$$\Delta G_t = kRT \log \left(10^{-1.0 - (-7.0)} \right)$$

$$\Delta G_t = kRT \log(10^{6.0}) = 6.0kRT$$

Therefore, ΔG_t is **proportional to the change in the pH** across the membrane.

(Choices A and C) Dividing the exponential expressions does not produce values of either $10^{(7/1)}$ or $10^{(1/7)}$, which would

be required to yield results of 7 or 1/7 from the logarithmic term of the equation.

(Choice D) When **dividing exponential expressions** of the same base (10, in this case), *exponents* are *subtracted* (numerator minus denominator); they are not added together. Elimination of the negative signs from the exponents must also result from the exponent subtraction operation, not by direct cancellation from the ratio.

Educational objective:

For a system at equilibrium, the change in the standard Gibbs free energy ΔG° is given by the equation $\Delta G^\circ = -RT \ln K_{\text{eq}}$, where R is the gas constant, T is the absolute temperature, and K_{eq} is the equilibrium constant. When a system is not at equilibrium, additional contributions to the Gibbs free energy must be evaluated based on the reaction quotient (concentration ratio) Q of the system.

Time spent: 0 sec

Subject:
General Chemistry

QID: 401573

System:
Solutions and Electrochemistry

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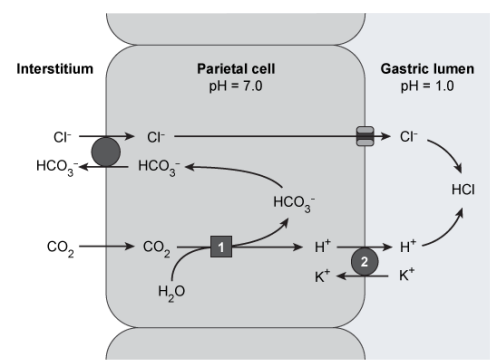


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Lactic	H ₃ C ₃ H ₅ O ₃	HC ₃ H ₅ O ₃ ⁻	1.4×10^{-4}
Ascorbic	H ₂ C ₆ H ₆ O ₆	HC ₆ H ₆ O ₆ ⁻	7.9×10^{-5}
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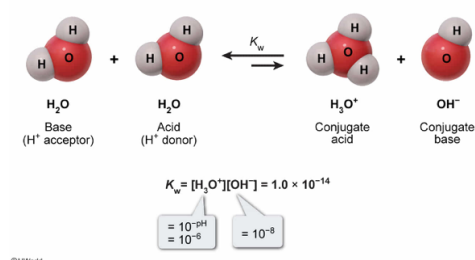
What is the approximate molar concentration of hydroxide in the duodenum (pH 6.0), where neutralization of the gastric acid leaving the stomach begins to take place? (Note: Assume that a sample is analyzed at 25 °C.)

- ☐ A. 1.0×10^{-14} M
- ✓ ☒ B. 1.0×10^{-8} M [78%]
- ☐ C. 1.0×10^{-7} M [6%]
- ☐ D. 1.0×10^{-6} M [14%]

Correct

77% answered correctly

Explanation:



A **pH** value correlates directly to $[\text{H}^+]$ by the relationship $[\text{H}^+] = 10^{-\text{pH}}$. The corresponding hydroxide concentration $[\text{OH}^-]$ is related to the $[\text{H}^+]$ concentration by the **self-ionization constant** K_w of water.

Although K_w **varies with temperature**, $K_w = 1.0 \times 10^{-14}$ at 25 °C and is therefore a useful standard for measurements.

$$K_w = [\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14} \text{ at } 25^\circ\text{C}$$

If the pH of the sample from the duodenum is 6.0, then the concentration of H^+ ions is:

$$[\text{H}^+] = 10^{-\text{pH}} = 10^{-6.0} = 1.0 \times 10^{-6}$$

Accordingly, **dividing** K_w at 25 °C by the known $[\text{H}^+]$ permits the calculation of $[\text{OH}^-]$:

$$[\text{OH}^-] = \frac{K_w}{[\text{H}^+]} = \frac{1.0 \times 10^{-14}}{1.0 \times 10^{-6}}$$

$$[\text{OH}^-] = 1.0 \times 10^{-14 - (-6)} = 1.0 \times 10^{-8} \text{ M}$$

Alternatively, the **pOH** can be used to find $[\text{OH}^-]$, given that pOH is defined as:

$$\text{pOH} = -\log [\text{OH}^-]$$

Applying the definition of pOH with a **logarithmic approach** to the K_w expression yields the relationship:

$$\text{pH} + \text{pOH} = 14$$

Calculation of the pOH can then be used to determine $[\text{OH}^-]$:

$$\text{pOH} = 14 - \text{pH}$$

$$\text{pOH} = 14 - 6.0 = 8.0$$

$$[\text{OH}^-] = 10^{-\text{pOH}} = 10^{-8.0} = 1.0 \times 10^{-8} \text{ M}$$

(Choice A) The value 1.0×10^{-14} is the self-ionization constant K_w rather than $[\text{OH}^-]$.

(Choice C) Based on the ionization constant K_w , the product of $[\text{OH}^-]$ and $[\text{H}^+]$ equals 10^{-14} . Therefore, a solution with an $[\text{OH}^-]$ of 1.0×10^{-7} M would contain an equal amount of $[\text{H}^+]$ and be completely neutral (pH 7.0). This is not the case, as the duodenum has pH 6.0.

(Choice D) The value 1.0×10^{-6} M is the $[\text{H}^+]$ (based on $\text{pH} = -\log[\text{H}^+]$ for pH 6.0); this is not the $[\text{OH}^-]$.

Educational objective:

The $[\text{H}^+]$ and $[\text{OH}^-]$ concentrations are related by the self-ionization constant K_w of water according to the expression $K_w = [\text{H}^+][\text{OH}^-]$, which equals 1.0×10^{-14} at 25 °C. Using a logarithmic approach, pH is related to $[\text{OH}^-]$ by the relationship $\text{pH} + \text{pOH} = 14$, where $\text{pOH} = -\log[\text{OH}^-]$.

Time spent:0 sec

Subject:
General Chemistry

QID:401574

System:
Solutions and Electrochemistry

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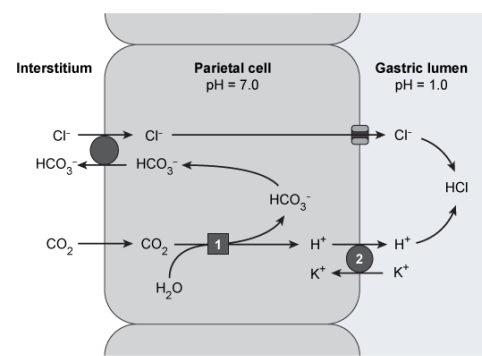


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Hydrochloric	HCl	Cl ⁻	1.3 × 10 ⁶
Carbonic	H ₂ CO ₃	HCO ₃ ⁻	4.3 × 10 ⁻⁷
Formic	HCO ₂ H	CO ₂ H ⁻	1.8 × 10 ⁻⁴
Acetic	HC ₂ H ₃ O ₂	C ₂ H ₃ O ₂ ⁻	1.8 × 10 ⁻⁵
Lactic	H ₃ C ₃ H ₅ O ₃	HC ₃ H ₅ O ₃ ⁻	1.4 × 10 ⁻⁴
Ascorbic	H ₂ C ₆ H ₆ O ₆	HC ₆ H ₆ O ₆ ⁻	7.9 × 10 ⁻⁵
Citric	H ₃ C ₆ H ₇ O ₇	H ₂ C ₆ H ₇ O ₇ ⁻	8.4 × 10 ⁻⁴

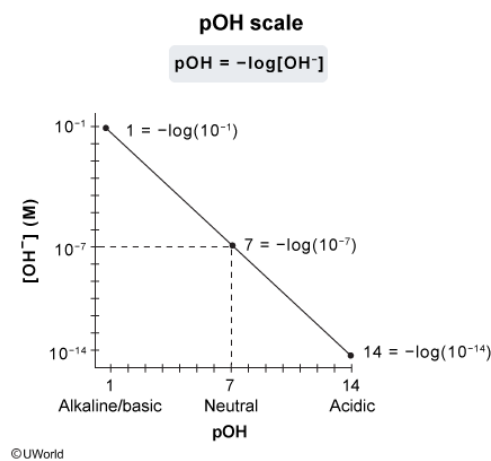
What happens to the pOH of the gastric acid entering the duodenum (pH 6.0) as a result of the introduction of bicarbonate ions from incoming pancreatic secretions?

- ✓ ☒ A. The pOH decreases as $[\text{OH}^-]$ increases. [48%]
☐ B. The pOH increases as $[\text{OH}^-]$ increases. [27%]
☐ C. The pOH increases as $[\text{OH}^-]$ decreases. [14%]
☐ D. The pOH decreases as $[\text{OH}^-]$ decreases. [9%]

Correct

48% answered correctly

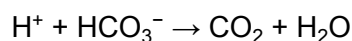
Explanation:



Just as the pH scale assesses acid-base measurements with respect to the concentration of the hydrogen ion $[\text{H}^+]$, the **pOH scale** assesses acid-base measurements with respect to the concentration of the **hydroxide ion $[\text{OH}^-]$** . Therefore, the pOH scale is **defined analogously to pH** as given by **$\text{pOH} = -\log[\text{OH}^-]$** .

Accordingly, lower pOH values indicate more alkaline solutions with higher $[\text{OH}^-]$. This is opposite the **pH scale**, in which lower pH values indicate more *acidic* solutions. Consequently, **pH and pOH are inversely correlated**: As pH rises, pOH falls (and vice versa) while maintaining the relationship $\text{pH} + \text{pOH} = 14$.

The bicarbonate ion (HCO_3^-) is a base (proton acceptor) capable of neutralizing gastric acid through the formation of carbon dioxide and water:



The $[\text{H}^+]$ decreases as bicarbonate consumes the available H^+ from the gastric acid, and the relative $[\text{OH}^-]$ increases. As a result, the duodenum (pH 6.0) is less acidic than the upper stomach (pH 1.0). Therefore, going from the highly acidic stomach to the less acidic duodenum gives a higher pH and a **lower pOH**, meaning $[\text{H}^+]$ decreases and $[\text{OH}^-]$ **increases**.

(Choices B, C, and D) Pancreatic secretions release bicarbonate (a base), which neutralizes gastric acid and causes an *increase* in $[\text{OH}^-]$. Increasing values of $[\text{OH}^-]$ yield *decreasing* values of pOH on the logarithmic pOH scale.

Educational objective:

Analogous to the pH scale, the pOH scale is defined as $\text{pOH} = -\log[\text{OH}^-]$. The pOH scale assesses acid-base measurements with respect to $[\text{OH}^-]$ whereas the pH scale operates with respect to $[\text{H}^+]$. The pH

and pOH scales are inversely correlated (ie, as one decreases, the other increases) such that $\text{pH} + \text{pOH} = 14$.

Time spent:0 sec

Subject:
General Chemistry

QID:401575

System:
Solutions and Electrochemistry

Gastric acid is one of many acids important to biological function in the human body. Production of the main component of gastric acid, hydrochloric acid (HCl), is accomplished in the stomach by the parietal cells, as shown in Figure 1.

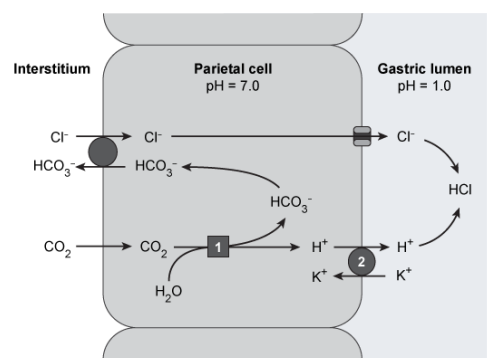


Figure 1 Formation of gastric acid in the stomach

Facilitated by the carbonate dehydratase enzyme (1), the parietal cells utilize the interaction between aqueous carbon dioxide and water in the blood as a source of bicarbonate ions and acidic H⁺ ions (protons). The H⁺ ions produced in this process are then exchanged for potassium ions through the cell membrane by a proton pump, the H⁺/K⁺-ATPase (2). In conjunction with this proton exchange, bicarbonate ions are also exchanged for chloride ions from the interstitial fluid. As a result, H⁺ and Cl⁻ ions are concentrated in the lumen of the stomach to form aqueous hydrochloric acid.

Because the production of the acid proceeds with a drop in pH from 7.0 in the cell cytosol to 1.0 in the stomach lumen, the process occurs against a concentration gradient and is endergonic, requiring the input of energy supplied by the hydrolysis of ATP. Ignoring effects from ion charge, the free energy change ΔG_t for ion transport across the cell membrane can be estimated by Equation 1:

$$\Delta G_t = kRT \log\left(\frac{C_f}{C_i}\right)$$

Equation 1

where R is the gas constant (8.32 J/mol·K), k is a logarithm conversion constant, T is the absolute temperature, and C_i and C_f are the initial and final molar concentrations across the gradient, respectively.

Although dilute, the gastric (hydrochloric) acid released into the lumen yields a strongly acidic environment crucial to protein digestion and the destruction of harmful micro-organisms. A comparison of its acid dissociation constant K_a to those of other biologically relevant acids is given in Table 1.

Table 1 Comparison of Strengths of Selected 0.10 M Acids at 25°C

Acid	Acid Formula (HA)	Conjugate Base (A ⁻)	K_a
Hydrochloric	HCl	Cl ⁻	1.3×10^6
Carbonic	H ₂ CO ₃	HCO ₃ ⁻	4.3×10^{-7}
Formic	HCO ₂ H	CO ₂ H ⁻	1.8×10^{-4}
Acetic	HC ₂ H ₃ O ₂	C ₂ H ₃ O ₂ ⁻	1.8×10^{-5}
Lactic	H ₃ C ₃ H ₅ O ₃	HC ₃ H ₅ O ₃ ⁻	1.4×10^{-4}
Ascorbic	H ₂ C ₆ H ₆ O ₆	HC ₆ H ₆ O ₆ ⁻	7.9×10^{-5}
Citric	H ₃ C ₆ H ₅ O ₇	H ₂ C ₆ H ₅ O ₇ ⁻	8.4×10^{-4}

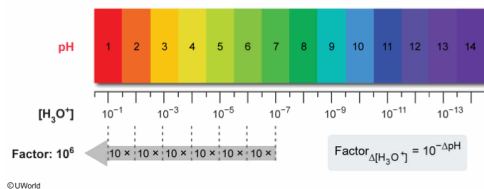
By what factor is the hydronium ion concentration increased in the stomach lumen compared to the parietal cell?

- ☐ A. 10^{-6} [17%]
☐ B. 10^{-1} [2%]
☐ C. 6 [4%]
✓ ☒ D. 10^6 [75%]

Correct

76% answered correctly

Explanation:



The pH scale is related to the concentration of the **hydronium ion** (H_3O^+) by the equation:

$$\text{pH} = -\log[\text{H}_3\text{O}^+]$$

This equation is often written in terms of hydrogen ion (H^+) concentration as $\text{pH} = -\log[\text{H}^+]$. The terms $[\text{H}^+]$ and $[\text{H}_3\text{O}^+]$ express the same meaning here (acid concentration) because acidic hydrogen ions are not free in aqueous solution but are carried by water molecules in the form of **hydronium ions** (H_3O^+).

Because the pH scale is logarithmic, the magnitude of **$[\text{H}_3\text{O}^+]$ changes** by a **power of 10** for **each pH unit**. The factor by which $[\text{H}_3\text{O}^+]$ changes has an exponential relationship to the change in pH (ie, the difference between the final pH and initial pH) as given by:

$$\text{Factor } \Delta[\text{H}_3\text{O}^+] = 10^{-(\text{final pH} - \text{initial pH})}$$

Moving from the parietal cell (pH 7.0) to the stomach lumen (pH 1.0) involves a change of 6 pH units. The increase in $[\text{H}_3\text{O}^+]$ going from the parietal cell to the stomach lumen can be determined as:

$$\text{Factor } \Delta[\text{H}_3\text{O}^+] = 10^{-(1 - 7)} = 10^6$$

Therefore, the $[\text{H}_3\text{O}^+]$ in the stomach lumen is one million (10^6) times stronger than the $[\text{H}_3\text{O}^+]$ in the cell cytosol.

(Choice A) 10^{-6} is the factor by which $[\text{H}_3\text{O}^+]$ changes when moving in the *reverse* direction to that stated in the question. The $[\text{H}_3\text{O}^+]$ in the cell cytosol is one millionth (10^{-6}) that of the $[\text{H}_3\text{O}^+]$ in the stomach lumen.

(Choice B) 10^{-1} is the $[\text{H}_3\text{O}^+]$ in the stomach lumen rather than the factor by which the concentration is increased relative to the $[\text{H}_3\text{O}^+]$ in the parietal cell.

(Choice C) 6 is the *difference* in pH, not the *factor* by which $[\text{H}_3\text{O}^+]$ is increased. Because pH is a

logarithmic scale, the $[\text{H}_3\text{O}^+]$ changes by a power of 10 for *each* pH unit.

Educational objective:

The pH unit is based on a logarithmic scale defined as $\text{pH} = -\log[\text{H}_3\text{O}^+]$. The magnitude of $[\text{H}_3\text{O}^+]$ changes by a power of 10 for each pH unit and is given by the expression: Factor $\Delta[\text{H}_3\text{O}^+] = 10^{-\Delta\text{pH}}$.

Time spent:0 sec

Subject:
General Chemistry

QID:401576

System:
Solutions and Electrochemistry